F23 Econ 284 Exam 1 Name:

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Each question is worth 10 points. The maximum length of your answer is given in the parentheses. Answer only numbered questions and answer in your own words. Answers similar to previous keys or online definitions will receive no credit.

The wage2 data set comes from the 1980 National Longitudinal Survey and includes the following variables:

wage	monthly earnings in $
educ	years of education
exper	years in profession
meduc	mother's education in years
feduc	father's education in years
urban	=1 if live in urban area

I ran the following regression

```
Call:
lm(formula = wage ~ educ + exper + meduc + feduc + urban, data = wage2)

Residuals:
    Min      1Q  Median      3Q     Max
-772.90 -241.96  -39.29  190.10 2075.30

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  -244.622    181.498  -1.348 0.178152
educ           47.364     12.722   3.723 0.000212 ***
exper          19.864      3.619   5.488 5.64e-08 ***
meduc          11.248     10.066   1.117 0.264111
feduc          10.645      9.348   1.138 0.255929
urban          86.342     43.879   1.985 0.044944 *
---
Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 369.4 on 715 degrees of freedom
  (213 observations deleted due to missingness)
Multiple R-squared:  0.1873,  Adjusted R-squared:  0.1805
F-statistic: 27.47 on 6 and 715 DF,  p-value: < 2.2e-16
```

1. Which of these results should we first check to determine whether or not we have a "good" model? Why? (two sentences)

The F-value and associated P-value because it reveals whether the model has statistically significant explanatory power. If this isn't the case, there is no reason to proceed.

2. Explain exactly what comparison the F-test represents in this regression and why it would be large if your model has significant explanatory power. (two sentences)

The F-test represents a comparison of the explanatory power (R-squared) of the model with all the regressors versus with no regressors. If this difference is not large, our regressors do not have significant explanatory power.

3. What do you conclude from the F-test? Discuss which numbers you use to come to your conclusion. (two sentences)

I conclude from the very small P-value that the regressors do have significant explanatory power.

4. Explain exactly what the Multiple R-squared represents in this case, using the variables in the model (ie, not just the generic "independent variables" and "dependent variables"). (one sentence)

The multiple r-squared value of 0.1873 means that years of education, years in profession, mother's education, father's education and whether or not a person lives in an urban area jointly explain 18.73% of the observed variation in monthly earnings.

5. What does a large t-value imply? Why is the estimate divided by its standard error a good measure of how far the estimate is from the hypothesized value, ie, why is the standard error a measure of "far"? (three sentences)

It implies that there is a significant relationship between the independent and dependent variable because the estimated coefficient that measures this relationship is statistically far from zero. The standard error is a measure of the expected random variation of the estimates. Thus, normalizing the difference between the estimate and the hypothesized value by dividing it by its standard error gives a measure of the distance above the random variation.

Interpret the coefficients, t-values and P values for the following variables. Be sure to interpret the numerical values of the coefficients. The words "unit" or "units" should not appear in your answer: use the actual units of the variables in the regression.

educ

6.1 Significance (one sentence):

At the alpha = 0.000212 level, we can reject the null hypothesis that education does not affect wage for a non-urban person.

6.2 Interpret coefficient value where appropriate (one sentence):

One year of additional education for a non-urban person is associated with a $47.364 higher monthly earnings.

meduc

7.1 Significance (one sentence):

It is not statistically significant, with a p-value of only 0.264111, and we fail to reject the null hypothesis that mother's education does not affect wage.

7.2 Interpret the coefficient value where appropriate (one sentence):

We do not have evidence that it is statistically significantly different from 0.

urban

8.1 Significance (one sentence):

At the alpha= 0.044944 level, we can reject the null hypothesis that living in an urban area does not affect wage.

8.2 Interpret the coefficient value where appropriate (one sentence):

Those living in an urban area earn, on average, $86.342 more per month than those not living in an urban area.

9.1 Is it surprising that meduc and feduc aren't significant? What might be the problem?

Yes, because we expect parents' education to be related to their income which generally affects their children's income through a variety of channels including the possibility of sending them to better schools, providing them with tutors, being able to give them more personal attention, etc.

9.2 How would you confirm that this is indeed a problem? (two sentences)

An important part would be seeing if they are correlated with one another. This could show if their co-movement hides their individual effects. Dropping one of the possibly correlated variables to see if the other becomes significant is another way to confirm.

I ran the following F-test.

```
> anova(UR,R)
Analysis of Variance Table

Model 1: wage ~ educ + exper + meduc + feduc + urban
Model 2: wage ~ educ + exper + urban
  Res.Df      RSS Df Sum of Sq     F    Pr(>F)
1    656 87731057
2    658 89958035 -2  -2226979 8.326 0.0002687 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

9.3 How do you interpret these results? What do you conclude? (two sentences)

These results indicate that our model lost significant explanatory power when we removed the parents' education variables. I conclude that these variables significantly affect their child's earnings and that multicollinearity was why their individual effects weren't significant in the model.

9.4 When we remove either meduc or feduc, the other variable (feduc or meduc) becomes significant. Is this consistent with multicollinearity? Explain. (two sentences)

Yes it is. It suggests that both significantly affect monthly wages, but their correlation results in us not being able to identify their individual effects when both are in the equation.

9.5 Why might removing either be a bad idea? Be very specific, including discussing how the two relevant conditions might be satisfied. (three sentences)

Removing either is a bad idea because each variable, when excluded, meets the conditions for OVB: 1. It is a determinant of earnings, and 2. It is correlated with the included variable, the other parent's education level.

10. 1 Person # 16 has 16 years of education, 17 years of experience, lives in an non-urban area and their parents each have 12 years of education. What is this person's predicted income from our regression?

Predicted earnings = -244.622 + 47.364*16+19.864*17+11.248*12+ 10.645*12= $1113.606

10.2 Person #16's actual monthly wage is $2137. What is the residual for this individual?

$2137-$1113.606 =$1023.394

10.3 Give examples of variables that might be in this residual.

Parent's income, ranking of university they attended, field of work, age, sex, race.

I reran the regression without using the robust=T command.

```
Call:
lm(formula = wage ~ educ + exper + meduc +
    feduc + urban, data = wage2)

Residuals:
    Min      1Q  Median      3Q     Max
-792.90 -228.69  -38.36  180.57 2059.72

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  -244.622    101.676  -2.405 0.021507 *
educ           47.364      9.228   5.132 0.000663 ***
exper          19.864      2.755   7.210 3.5e-07  ***
meduc          11.248      6.274   1.793 0.097527 .
feduc          10.645      5.103   2.086 0.037398 *
urban          86.342     45.575   1.894 0.067990 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 365.7 on 656 degrees of freedom
Multiple R-squared:  0.198,   Adjusted R-squared:  0.1907
F-statistic:    27 on 6 and 656 DF,  p-value: < 2.2e-16
```

And here are the original results, copied here for your convenience.

```
Call:
```

```
lm(formula = wage ~ educ + exper + meduc + feduc + urban, data = wage2)

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```

11. What do these two sets of results tell us? Explain (two sentences)

The change in the standard errors tell us that we had a heteroscedasticity problem in our data and that running robust=T was essential to getting reliable results.